

網路系統統整與實作 作業六

1. **Demo time:** 2026/6/16 or 2026/6/17 (我請助教安排)
2. **學習目標:** 使用開源軟體，建置 5G 行動通訊系統。使用的軟體包含:
 - (1) Open5GS
 - (2) srsRAN
3. **Part A — Open5GS Installation (15%)**
 - (1) Install MongoDB.
 - (2) Install Open5GS.
 - (3) Verify service status (must be “active”).
 - (4) Add a subscriber.
4. **Part B — srsRAN Installation (15%)**
 - (1) Install srsRAN.
 - (2) Launch gNB.
5. **Part C — ZMQ Virtual Radio Deployment (15%)**
 - (1) Please successfully establish a PDU session.
6. **Part D — Wireshark Signaling Analysis (15%)**
 - (1) Please capture traffic on lo & n3
 - (2) Identify messages below
 - Registration request
 - Authentication request
 - Authentication response
 - Security mode command
 - Registration accept
 - (3) Please draw a signaling flow diagram and explain the purpose.
7. **Part E — Configuration Tuning Experiments (20%)**
 - (1) PLMN configuration
 - Original configuration
 - mcc: 001
 - mnc: 01
 - Modified configuration
 - mcc: 524
 - mnc: 98
 - Does UE registration succeed? Why or why not?
 - (2) Tracking Area Code (TAC)
 - Original configuration
 - tac: 7

- Modified configuration
 - tac: 100
 - How does the UE behave after the change? Why is TAC important in cellular networks?
- (3) AMF Address Configuration
- Original configuration
 - amf address: 127.0.0.5
 - Modified configuration
 - amf address: 10.10.10.10
 - Suppose the NG setup would fail. How would you troubleshoot this issue? Which logs should be examined?
 - How does the UE behave after the change? Why is TAC important in cellular networks?

8. Part F — Performance Evaluation (10%)

- (1) Bandwidth configuration
- Please compare the settings of 10 MHz and 20 MHz.
 - How does bandwidth affect throughput?
 - Is the relationship linear?
- (2) MCS configuration
- Original configuration
 - max_mcs: 10
 - Modified configuration
 - max_mcs: 28
 - How does MCS affect throughput?
 - What are the trade-offs of using a higher MCS?
- (3) Scheduler comparison
- Please compare round robin scheduler and proportional fair scheduler.
 - Evaluation metrics are “throughput”, “fairness”, and “resource utilization”.
 - Which scheduler provides higher throughput?
 - Which scheduler provides better fairness?

9. Part G — 5G Network Slicing Configuration and Evaluation (10%)

- (1) Configure multiple slices in Open5GS
- eMBB slice
 - SST=1, SD=000001
 - URLLC slice
 - SST=2, SD=000002
- (2) Configure subscriber profiles: create two subscribers
- Subscriber A
 - IMSI: 001010000000001
 - Slice: SST=1, SD=000001
 - Service: video streaming
 - Subscriber B

- IMSI: 0010100000000001
- Slice: SST=2, SD=000002
- Service: industrial control

(3) Configure Slice information in srsRAN: modify gNB configuration to support both slice types.

(4) Traffic Differentiation: Generate two traffic types.

- eMBB: high-throughput traffic
- URLLC: latency-sensitive traffic

(5) Performance measurements for eMBB and URLLC slices

- Throughput
- Packet loss
- RTT

(6) Congestion Experiment

- Please create congestion.

(7) Wireshark Analysis

- Please capture “NGAP” and “NAS” messages.
- Please identify “Allowed NSSAI”, “Configured NSSAI”, and “Requested NSSAI”.

10. Bonus Lab: USRP B210 OTA Deployment (10%)

(1) Objective: deploy a real over-the-air 5G network using a USRP B210 SDR.

(2) Tasks: use a smartphone to connect to the private 5G network.

(3) What to measure?

- RSRP
- RSRQ
- SINR

(4) Please adjust TX gain to show the changes of RSRP.